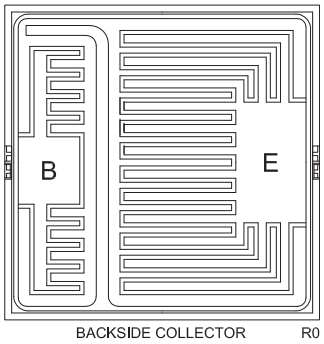
CP527-2N6299

PNP - Darlington Transistor Die
8.0 Amp, 80 Volt

The CP527-2N6299 is a silicon PNP Darlington power transistor designed for high gain amplifier applications.



MECHANICAL SPECIFICATIONS:

Die Size	110 x 110 MILS
Die Thickness	10.6 MILS
Base Bonding Pad Size	21 x 24 MILS
Emitter Bonding Pad Size	24 x 42 MILS
Top Side Metalization	Al – 20,000Å
Back Side Metalization	Ni/Ag – 2,000Å/10,000Å
Scribe Alley Width	4.3 mils
Wafer Diameter	4 INCHES
Gross Die Per Wafer	875

MAXIMUM RATINGS: (T_A=25°C)	**SYMBOL**		**UNITS**
Collector-Base Voltage	V_{CBO}	80	V
Collector-Emitter Voltage	V_{CEO}	80	V
Emitter-Base Voltage	V_{EBO}	5.0	V
Continuous Collector Current	I_C	8.0	A
Operating and Storage Junction Temperature	T_J, T_{stg}	-65 to +150	°C

ELECTRICAL CHARACTERISTICS: (T_A=25°C unless otherwise noted)

SYMBOL	**TEST CONDITIONS**	**MIN**	**MAX**	**UNITS**
I_{CEV}	V_{CE}=80V, V_{BE}=1.5V		0.5	mA
I_{CEO}	V_{CE}=40V		0.5	mA
I_{EBO}	V_{EB}=5.0V		2.0	mA
BV_{CEO}	I_C=100mA	80		V
$V_{CE(SAT)}$	I_C=4.0A, I_B=16mA		2.0	V
$V_{CE(SAT)}$	I_C=8.0A, I_B=80mA		3.0	V
$V_{BE(SAT)}$	I_C=8.0A, I_B=80mA		4.0	V
$V_{BE(ON)}$	V_{CE}=3.0V, I_C=4.0A		2.8	V
h_{FE}	V_{CE}=3.0V, I_C=4.0A	750	18K	
h_{FE}	V_{CE}=3.0V, I_C=8.0A	100		
h_{fe}	V_{CE}=3.0V, I_C=3.0A, f=1.0kHz	300		
f_T	V_{CE}=3.0V, I_C=3.0A, f=1.0MHz	4.0		MHz
C_{ob}	V_{CB}=10V, I_E=0, f=100kHz		300	pF

CP527-2N6299
Typical Electrical Characteristics

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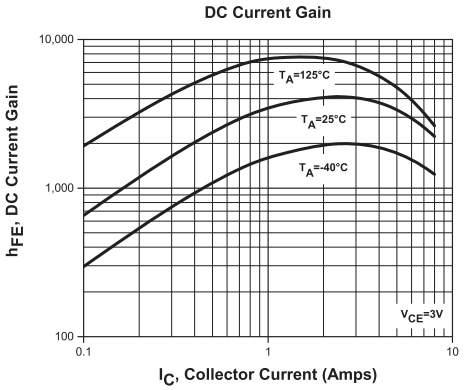

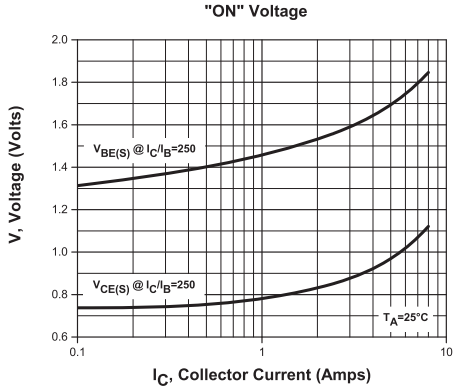

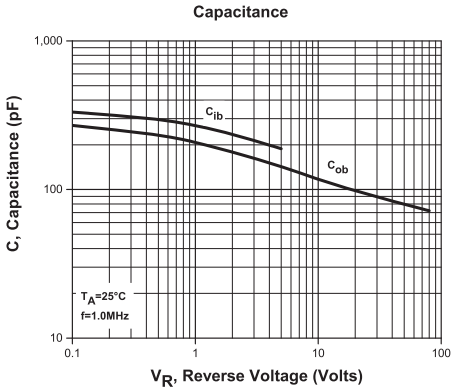

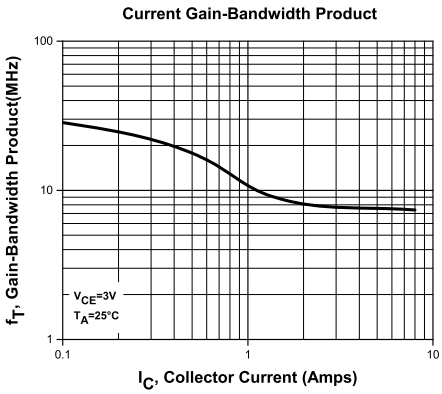

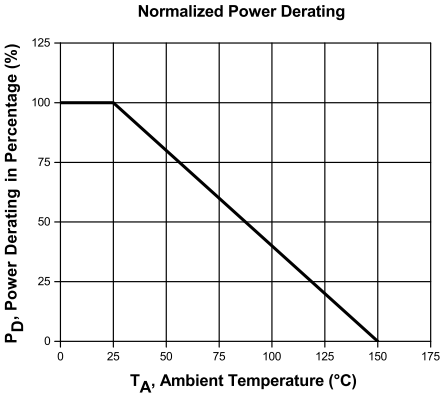

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